

Supplementary Materials for

**In-situ characterization and quantification of melt pool variation under constant input
energy density in laser powder-bed fusion additive manufacturing process**

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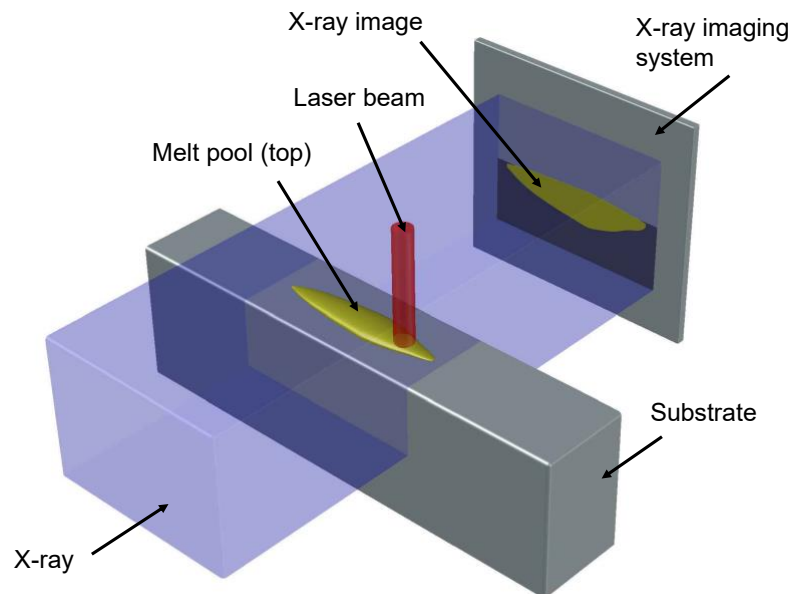
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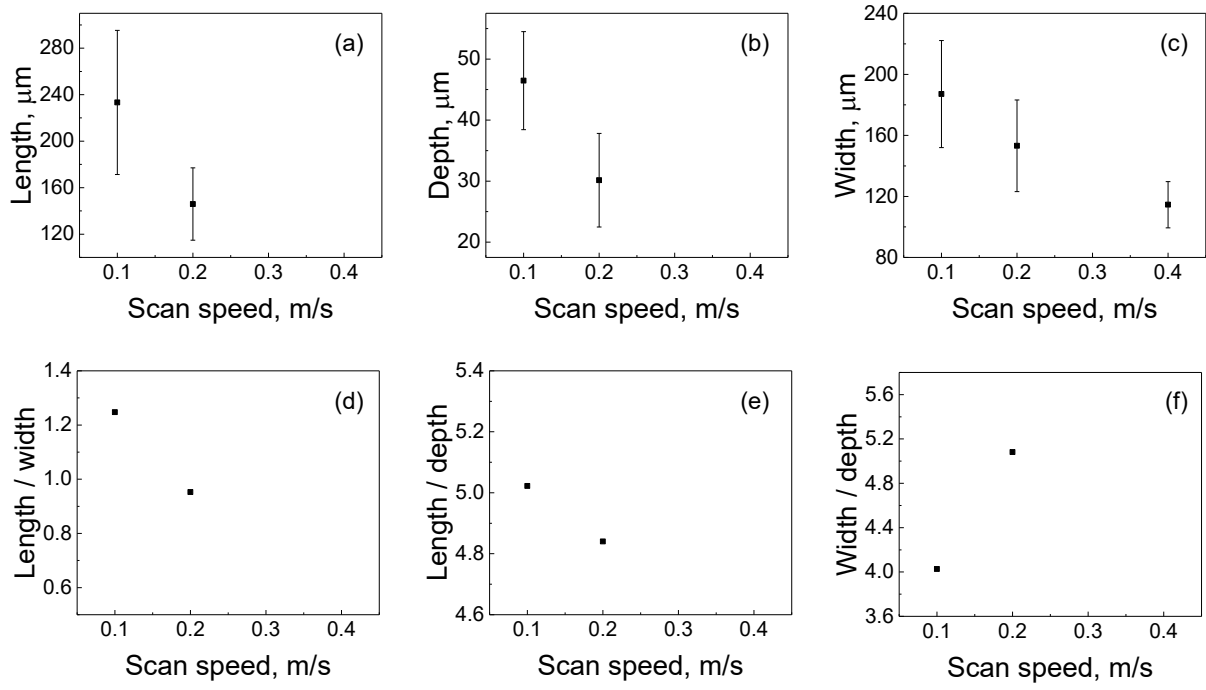
Supplementary Figures

Supplementary Figure 1



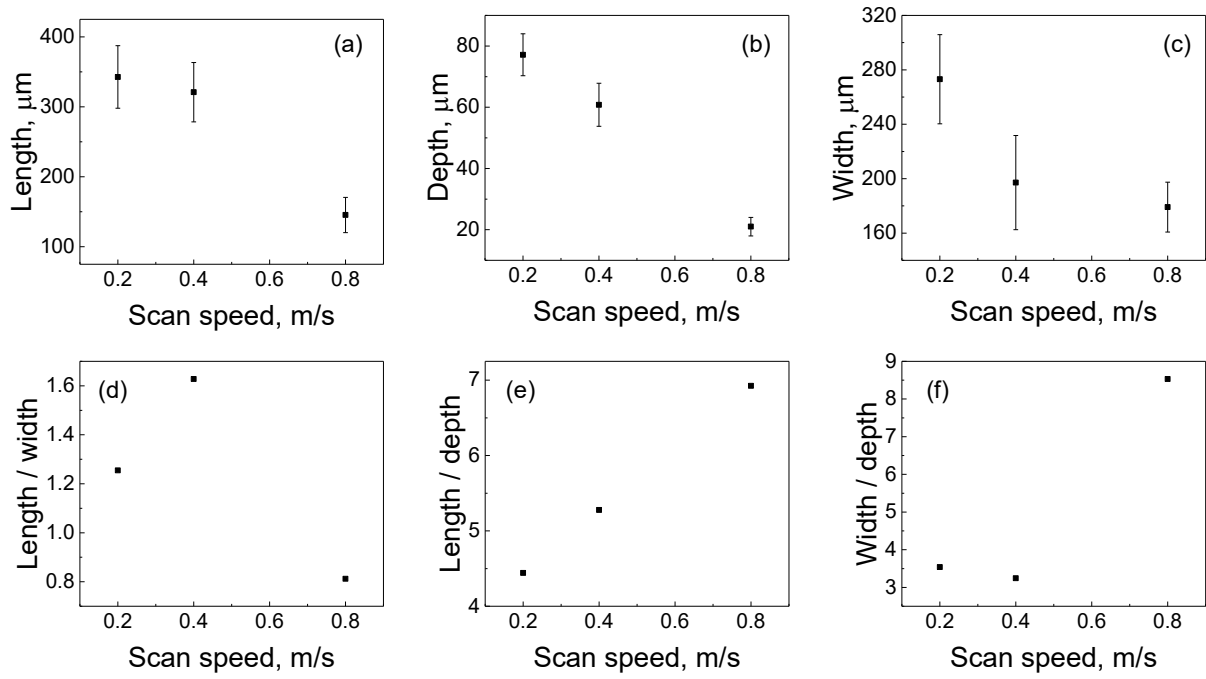
Supplementary Figure 1 | Schematic showing how the melt pool information can be projected into x-ray images. The powder bed is not shown in the schematic.

Supplementary Figure 2



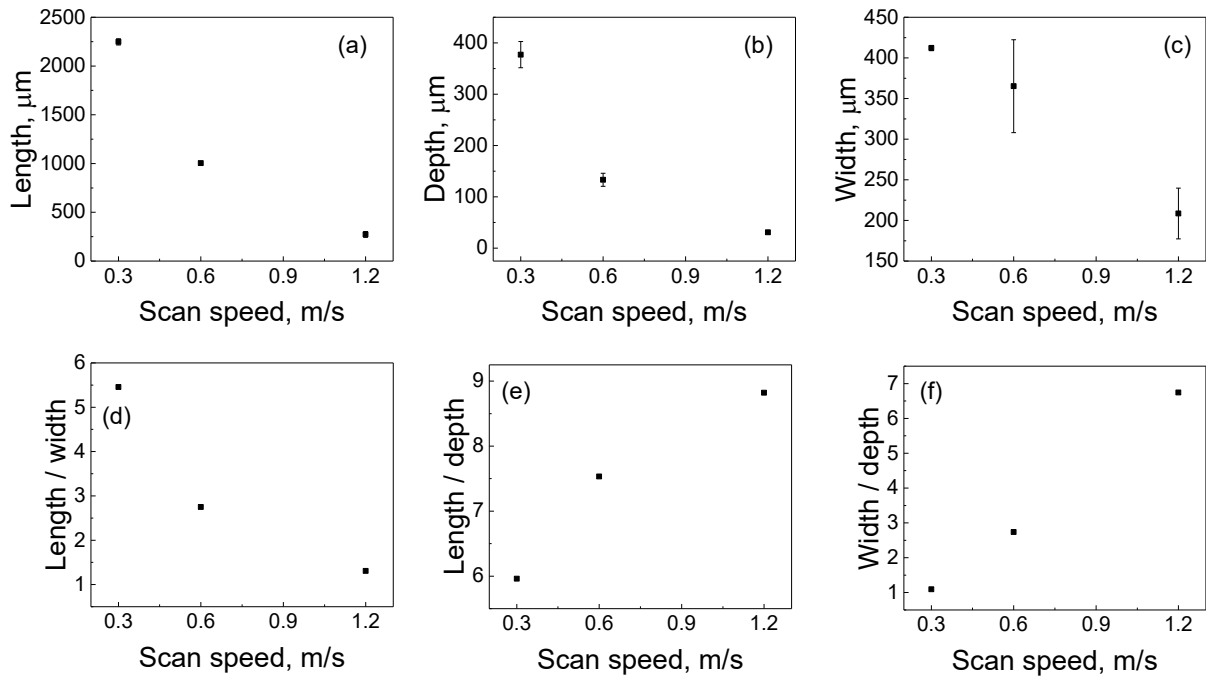
Supplementary Figure 2 | Melt pool dimensions and dimension ratios as functions of laser scan speed under a constant laser power of 104 W.

Supplementary Figure 3



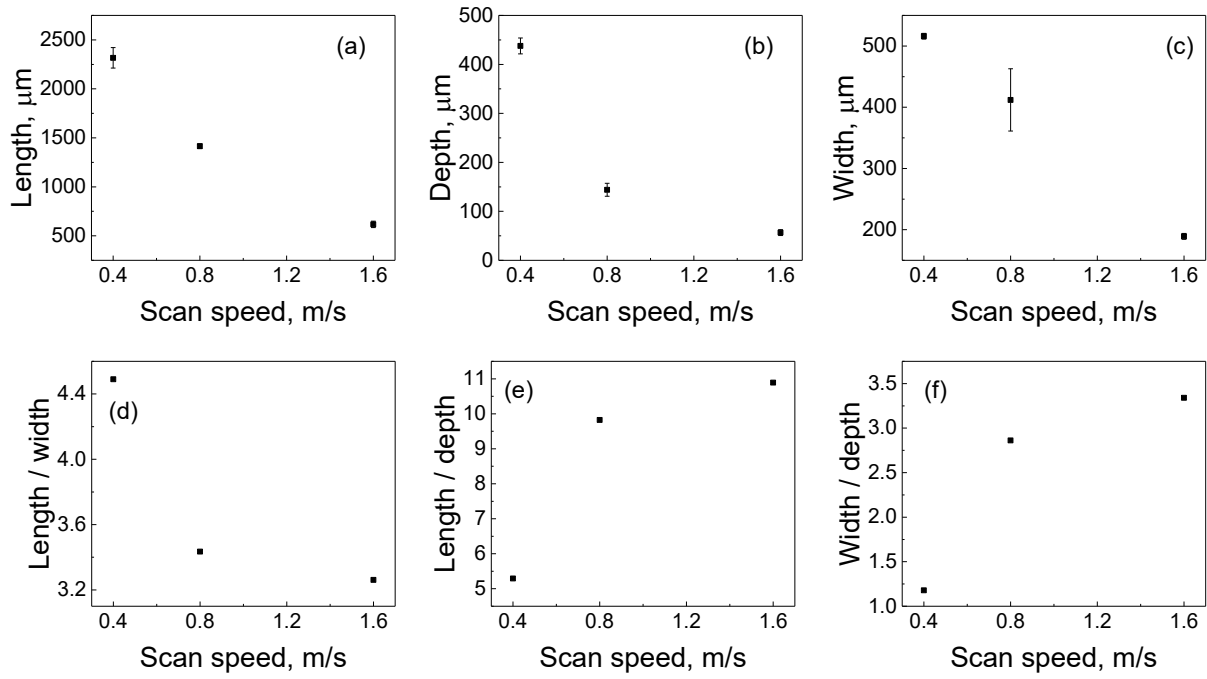
Supplementary Figure 3 | Melt pool dimensions and dimension ratios as functions of laser scan speed under a constant laser power of 208 W.

Supplementary Figure 4



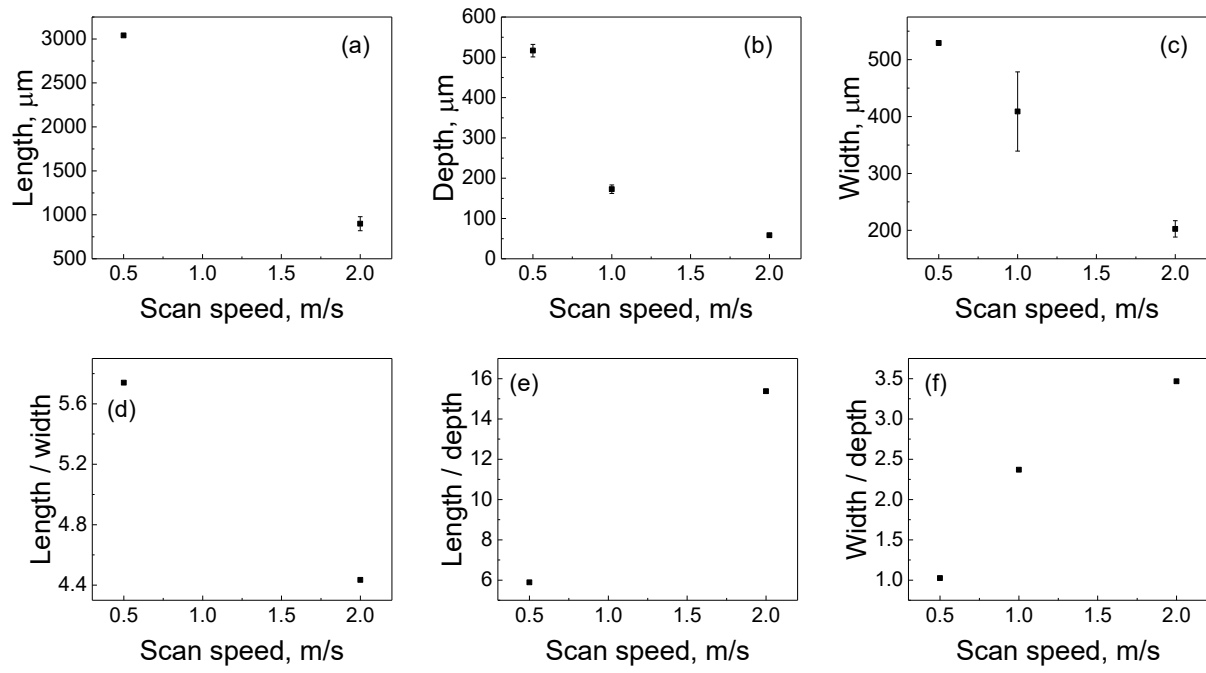
Supplementary Figure 4 | Melt pool dimensions and dimension ratios as functions of laser scan speed under a constant laser power of 312 W.

Supplementary Figure 5



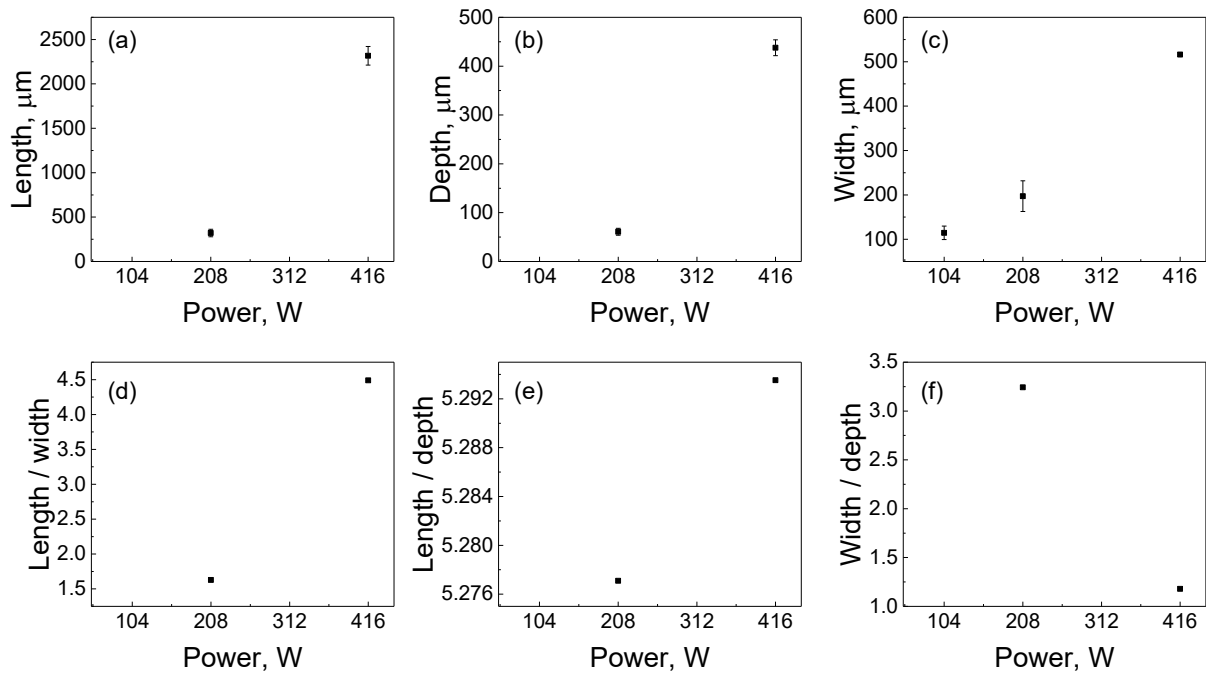
Supplementary Figure 5 | Melt pool dimensions and dimension ratios as functions of laser scan speed under a constant laser power of 416 W.

Supplementary Figure 6



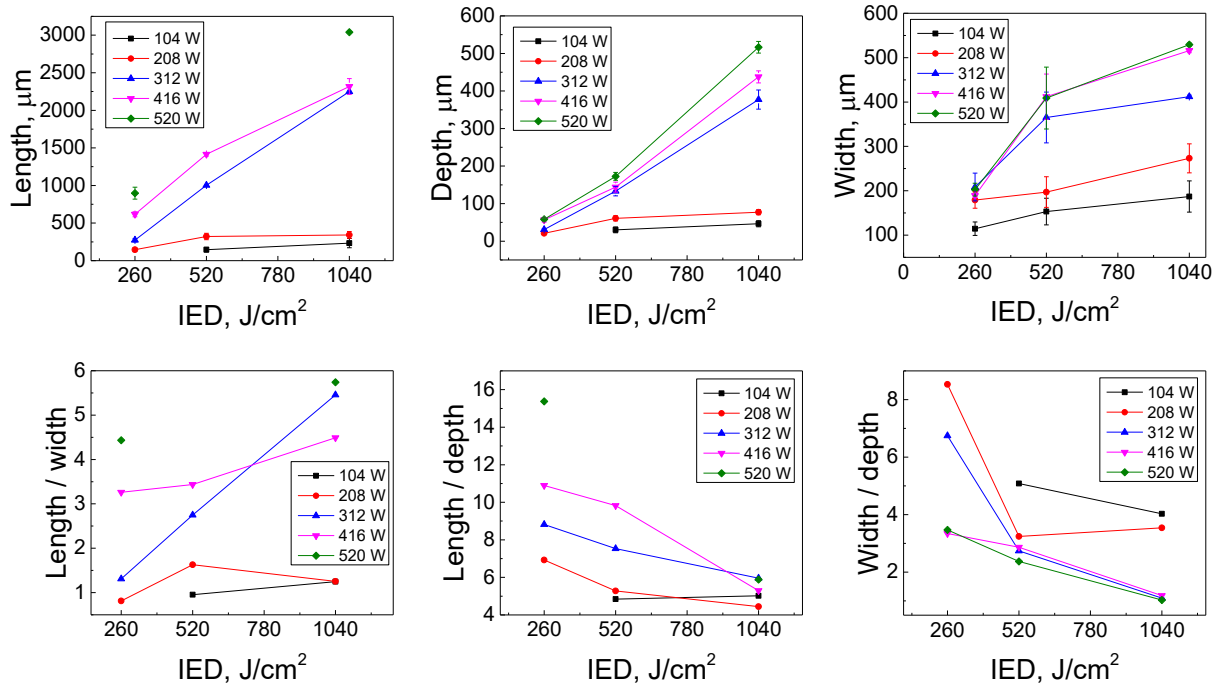
Supplementary Figure 6 | Melt pool dimensions and dimension ratios as functions of laser scan speed under a constant laser power of 520 W.

Supplementary Figure 7



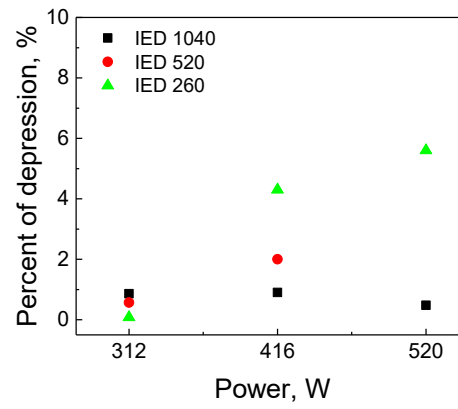
Supplementary Figure 7 | Melt pool dimensions and dimension ratios as functions of laser power under a constant laser scan speed of 0.4 m/s.

Supplementary Figure 8



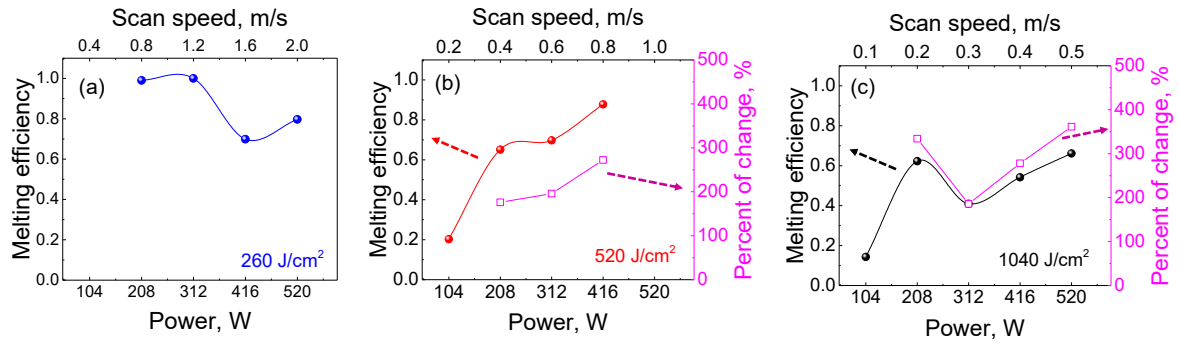
Supplementary Figure 8 | Melt pool dimensions and dimension ratios as functions of IED.

Supplementary Figure 9



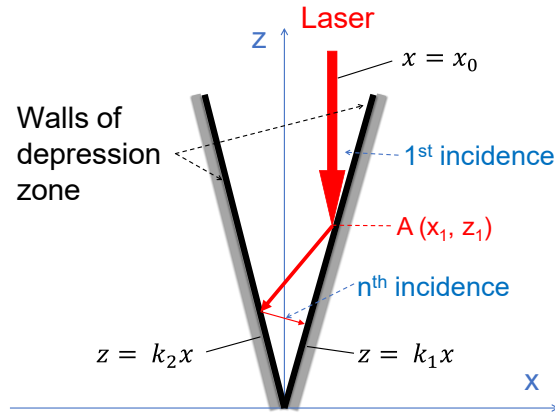
Supplementary Figure 9 | Percentage of depression zone volume over melt pool volume.

Supplementary Figure 10



Supplementary Figure 10 | Melting efficiency as a function of processing conditions under constant IED level. The melting efficiency is plotted as a function of laser power (bottom x-axis) and laser scan speed (top x-axis) under constant IED levels. The experiments were conducted under three IED levels: (a) 260 J/cm², (b) 520 J/cm², and (c) 1040 J/cm². The hollow squares show the percentage of change under each processing condition with respect to the condition of 104 W. The percentage of change is not calculated in (a) due to the lack of melt pool under 104 W.

Supplementary Figure 11



Supplementary Figure 11 | Schematic of ray-tracing model. The first laser incidence acts at the center of the front depression zone wall.

Supplementary Notes

Supplementary Note 1 | Melting efficiency variation under constant *IED* levels

Melting efficiency (η_m) describes the ratio of energy for melting the material (and heating up the molten material) over the total energy absorbed by the material, as expressed by:

$$\eta_m = \frac{E_m}{E_{ab}} \quad (\text{S-1})$$

where E_m is the energy required to melt the material, E_{ab} is the total energy absorbed by the material. As explained in the main text section 4.2, the energy absorption is mainly due to the Fresnel laser absorptivity (a_F). Therefore, equation (S-1) can be re-written into:

$$\eta_m = \frac{E_m}{E_{ab}} = \frac{E_m}{a_F \cdot P \cdot \tau} \quad (\text{S-2})$$

where P is the laser power, τ is the laser exposure time. Since the Fresnel laser absorptivity (a_F) has been calculated in the main text section 4.2, the key of calculating the melting efficiency is to estimate the energy required to melt the material (E_m). To generate a certain size of melt pool, an energy conservation equation can be built as:

$$E_m = m \cdot (\int C dT + L) \quad (\text{S-3})$$

where m is the mass of the melt pool, C is the specific heat of the material, L is the latent heat during solid-liquid phase transformation. Taking $m = V \cdot \rho_{liq}$, where V is the volume of the melt pool, ρ_{liq} is the density of the liquid metal; $\int C dT = \int_{T_o}^{T_m} C_s dT + \int_{T_m}^{\frac{T_m+T_b}{2}} C_l dT$, where T_o , T_m , T_b are the room temperature (20 °C), melting point, and boiling point of the material, C_s and C_l are the specific heat of solid and liquid material, equation (S-3) can be re-written as:

$$E_m = V \cdot \rho_{liq} \cdot \left(\int_{T_o}^{T_m} C_s dT + \int_{T_m}^{\frac{T_m+T_b}{2}} C_l dT + L \right) \quad (S-4)$$

For AlSi10Mg alloy, all of the material physical properties are published in reference [1]: $T_m = 555$ °C, $T_b = 2467$ °C, $L = 476.5$ kJ/kg, $C_s = 1098$ J/(kg·°C), $C_l = 1175$ J/(kg·°C), and $\rho_{liq} = 2210$ kg/m³. The material process-related property, melt pool volume (V), has been calculated and presented in the main text section 3.3. Substituting the above numbers into equation (S-4), the energy required to melt the material (E_m) can be obtained. Substituting E_m into equation (S-2), along with $\tau = L/v$ (where L is the length of melt pool, v is the laser scan speed), the equation of melting efficiency η_m can be solved. To be noted, since the upper limit of the integral $C_l dT$ is $(T_m+T_b)/2$, this term is overestimated for the case of elongated melt pool. Therefore, the calculated E_m is the upper limit of all possible conditions.

The results of melting efficiency under three constant IED levels are exhibited in **Supplementary Figure 10**. The maximum efficiency is around four times as compared to the lowest efficiency, under all conditions studied in the paper.

Supplementary Note 2 | Calculation of laser absorptivity through a ray-tracing model

For each of the interactions between the laser and keyhole wall, according to the Fresnel equation, the absorptivity is related to the incident angle θ . It can be calculated by the polarization form, as provided in the following [2]:

$$\text{TE: } r_{TE} = \frac{\cos \theta - \sqrt{(n_R^2 - n_I^2 - \sin^2 \theta) + i(2n_R n_I)}}{\cos \theta + \sqrt{(n_R^2 - n_I^2 - \sin^2 \theta) + i(2n_R n_I)}} \quad (S-5)$$

$$\text{TM: } r_{TM} = \frac{[n_R^2 - n_I^2 + i(2n_R n_I)] \cos \theta - \sqrt{(n_R^2 - n_I^2 - \sin^2 \theta) + i(2n_R n_I)}}{[n_R^2 - n_I^2 + i(2n_R n_I)] \cos \theta + \sqrt{(n_R^2 - n_I^2 - \sin^2 \theta) + i(2n_R n_I)}} \quad (S-6)$$

where r_{TE} is the reflection coefficient of the TE polarization mode, r_{TM} is the reflection coefficient of the TM polarization mode, n_R and n_I are the real part and imaginary part of the complex refractive index of molten aluminum, respectively. According to (S-5) and (S-6), the reflection coefficient for each polarization can be reduced to a ratio of complex numbers in the form of $(a + bi)/(c + di)$, so that

$$R_{TE} = r_{TE}^2 = (a_{TE}^2 + b_{TE}^2)/(c_{TE}^2 + d_{TE}^2) \quad (S-7)$$

$$R_{TM} = r_{TM}^2 = (a_{TM}^2 + b_{TM}^2)/(c_{TM}^2 + d_{TM}^2) \quad (S-8)$$

where R_{TE} and R_{TM} are the reflectance for TE mode and TM mode, respectively. Generally, the reflectance can be expressed as a combination of TE and TM:

$$R = (R_{TE} + R_{TM})/2 \quad (S-9)$$

Since the shape of the depression zone has been captured by the x-ray images, the reflection times and the incident angle for each reflection can be determined by tracing a ray of light to see how it propagates in the keyhole. The following assumptions were made to facilitate the ray-tracing process: (1) Assume the front and rear keyhole walls are straight; (2) Assume the incident beam first interact at the center of front keyhole wall; (3) Assume the incident beam comes from the vertical direction. (4) Assume the laser beam as a geometrical line. The beam intensity distribution along the radius was not considered. A schematic of this simplified model is displayed in **Supplementary Figure 11**. By fitting the captured position data of the keyhole front and rear walls, they can be expressed by the following functions:

For the front wall

$$z = k_1 x \quad (S-10)$$

For the rear wall

$$z = k_2 x \quad (S-11)$$

where k_1 and k_2 are the slopes of the front and rear keyhole walls.

The equation of the ray of the laser beam can be expressed as

$$x = x_0 \quad (\text{S-12})$$

The first incident point A ($x_0, k_1 x_0$) can be calculated by solving the simultaneous equations (S-10) and (S-12).

The slope of the normal line of the keyhole front wall at the first incident point is

$$k_{n1} = -\frac{1}{k_1} \quad (\text{S-13})$$

The equation for the normal line of the keyhole front wall at the first reflection can thus be obtained by the incident point A and normal line slope k_{n1} :

$$z = k_{n1}x + l_{n1} \quad (\text{S-14})$$

The expression of the reflected beam can then be obtained, as the incident beam and the reflected beam are axisymmetric with respect to the normal line of the front keyhole wall. Treating the reflected beam as the incident beam for the next reflection, the calculation will continue until the beam escapes from the depression zone.

With the incident angle from a ray tracing model and (S-9), the absorptivity of the keyhole wall at nth reflection can be calculated:

$$\alpha_n = [\prod_{j=1}^{n-1} R(j)][1 - R(n)] \quad (\text{S-15})$$

where $R(n)$ is the reflectance at the nth reflection. The overall absorption is the sum of absorptivity at each reflection point, as follows:

$$\alpha = \sum_1^n \alpha_n = 1 - \prod_1^n R(n) \quad (\text{S-16})$$

Supplementary references

- [1] S. Sulaiman, L.Y. Pio, Validation of Magmasoft simulation of the sand casting process, Inst. Eng. Malaysia. 65 (2004) 55-64.
- [2] F.L. Pedrotti, L.S. Pedrotti, Introduction to Optics (3rd Edition), Pearson Education. (2006) 656.